

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

593000501 - Embedded Platforms And Communications For Iot

DEGREE PROGRAMME

59AH - Master Universitario En Internet Of Things (iot)

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

Index

Learning guide

1. Description.....	1
2. Faculty.....	1
3. Prior knowledge recommended to take the subject.....	2
4. Skills and learning outcomes	3
5. Brief description of the subject and syllabus.....	4
6. Schedule.....	6
7. Activities and assessment criteria.....	8
8. Teaching resources.....	10
9. Other information.....	11

1. Description

1.1. Subject details

Name of the subject	593000501 - Embedded Platforms And Communications For Iot
No of credits	4.5 ECTS
Type	Compulsory
Academic year of the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	59AH - Master Universitario en Internet Of Things (Iot)
Centre	59 - E.T.S. De Ingeniería Y Sist. De Telecom.
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Miguel Chavarrias Lapastora	A4204	miguel.chavarrias@upm.es	Sin horario. Schedules are subject to possible changes. Please, confirm the latest version with the lecturer

Eduardo Juarez Martinez	A4204	eduardo.juarez@upm.es	Sin horario. Schedules are subject to possible changes. Please, confirm the latest version with the lecturer
Guillermo Azuara De Pablo (Subject coordinator)	A4206	g.azuara@upm.es	Sin horario. Schedules are subject to possible changes. Please, confirm the latest version with the lecturer.

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Application of processor interrupts
- Programming and debugging using the C language (with emphasis on structures, pointers and memory management)
- Application of the Von Neumann's computer architecture
- Application of processor peripherals

4. Skills and learning outcomes *

4.1. Skills to be learned

CB06 - Poseer y comprender conocimientos que aporten una base u oportunidad de ser originales en el desarrollo y/o aplicación de ideas, a menudo en un contexto de investigación

CB10 - Que los estudiantes posean las habilidades de aprendizaje que les permitan continuar estudiando de un modo que habrá de ser en gran medida autodirigido o autónomo.

CE.01 - Evaluar las arquitecturas HW/SW existentes y diseñar plataformas embebidas para desarrollar soluciones IoT integrando diversas capas SW que manejen sensores, actuadores y dispositivos de comunicación

CE.02 - Diseñar y desarrollar redes de sensores integrando nodos heterogéneos con diferentes sistemas de comunicación inalámbricas para desarrollar aplicaciones IoT

CG01 - Los alumnos demostrarán tener una visión del estado actual, las necesidades y los problemas que se plantean en el mundo de la IoT, así como de las arquitecturas y estándares más utilizados

CG02 - Los alumnos serán capaces de aplicar métodos y tecnologías avanzadas que les permitan abordar necesidades y problemas en aplicaciones IoT

CT.01 - Capacidad de uso de la lengua inglesa para el trabajo en contextos internacionales

4.2. Learning outcomes

RA24 - To integrate a wireless technology and the communication protocols for a hardware platform in an IoT specific application

RA22 - To combine the development tools for the integration of all software elements required to use a hardware platform in an IoT solution

RA21 - To establish the building or selection criteria of embedded hardware platforms for the integration of a specific IoT application

RA23 - To use the interfaces for the connection of sensors, actuators and communication modules in a hardware/software platform supporting IoT applications

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

Brief Description of the Course

The main goal of this course is to introduce the hardware and software architectures frequently used in hardware embedded platforms for IoT applications. More specifically, the description of the essential hardware resources, architecture details and sensor and actuator interfaces are covered. In addition, the integration, development and debugging of software IoT applications are studied in detail.

5.2. Syllabus

1. IoT Introduction
 - 1.1. General Introduction
 - 1.2. Technology Solutions
 - 1.3. Use-case Description
 - 1.4. Platform Architecture
 - 1.5. Demo
2. Zephyr OS/VS Code: the Software Perspective
 - 2.1. Zephyr OS & VS Code
 - 2.2. I/O with Zephyr OS
 - 2.3. Zephyr Operating System
 - 2.4. Hands-on Lab
3. Hw Architectures and I/O Interfaces: the Hardware Perspective
 - 3.1. Embedded Hardware for IoT
 - 3.2. Serial Interfaces: I2C
 - 3.3. Serial Interfaces: SPI

3.4. Serial Interfaces: RS-232

4. Project

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Lesson 1: General Introduction Duration: 02:00 Lecture Lesson 1: Technology solutions Duration: 01:00 Lecture Lesson 1: Use-case description Duration: 01:30 Lecture			
2	Lesson 1: Platform Architecture Duration: 03:00 Lecture	Lesson 1: Demo Duration: 01:00 Lecture		
3	Lesson 2: Zephyr Operating System Duration: 03:00 Lecture	Lesson 2: Zephyr & IDE Duration: 01:00 Laboratory assignments		
4	Lesson 2: I/O with Zephyr OS Duration: 02:00 Laboratory assignments Lesson 3: Embedded Hardware Description Duration: 02:00 Additional activities			
5		Lesson 2: Hands-on Lab Duration: 02:00 Laboratory assignments Hands-on exercises to assess proficiency Duration: 02:00 Additional activities		Lesson 2: Hands-on exercises to assess proficiency Group work Progressive assessment Presential Duration: 02:00
6		Lesson 3: Serial Interfaces: SPI Duration: 01:00 Laboratory assignments Lesson 3: Serial Interfaces: I2C Duration: 02:00 Laboratory assignments Lesson 3: Serial Interfaces: RS-232 Duration: 01:00 Laboratory assignments		

7	Lesson 4: Course Project Duration: 02:00 Additional activities			Lesson 4: Preliminary Report Assessment on Course Project Group work Progressive assessment Presential Duration: 02:00
8		Lesson 4: Course Project Duration: 04:00 Laboratory assignments		
9		Lesson 4: Course Project Duration: 02:00 Additional activities		Lesson 4: Project initial validation Group work Progressive assessment Presential Duration: 02:00
10		Lesson 4: Course Project Duration: 04:00 Laboratory assignments		
11		Project assessment Duration: 02:00 Additional activities		Lesson 4: Project assessment Problem-solving test Progressive assessment Presential Duration: 02:00 Lesson 4: Report assessment on course project Group work Progressive assessment Not Presential Duration: 01:00
12				
13				
14				
15				
16				
17				

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
5	Lesson 2: Hands-on exercises to assess proficiency	Group work	Face-to-face	02:00	15%	0 / 10	CB06 CB10 CG01 CG02 CT.01 CE.01
7	Lesson 4: Preliminary Report Assessment on Course Project	Group work	Face-to-face	02:00	15%	0 / 10	CB06 CB10 CG01 CG02 CT.01 CE.01 CE.02
9	Lesson 4: Project initial validation	Group work	Face-to-face	02:00	20%	0 / 10	CB06 CB10 CG01 CG02 CT.01 CE.01 CE.02
11	Lesson 4: Project assessment	Problem-solving test	Face-to-face	02:00	35%	0 / 10	CB06 CB10 CG01 CG02 CT.01 CE.01 CE.02
11	Lesson 4: Report assessment on course project	Group work	No Presential	01:00	15%	0 / 10	CB06 CB10 CG01 CG02 CT.01 CE.01 CE.02

7.1.2. Global examination

No se ha definido la evaluación sólo por prueba final.

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Course Project Exam	Other assessment	Face-to-face	03:00	80%	0 / 10	CB06 CB10 CG01 CG02 CT.01 CE.01 CE.02
Proposed Exercises Exam	Other assessment	Face-to-face	02:00	20%	0 / 10	CB06 CB10 CG01 CG02 CT.01 CE.01

7.2. Assessment criteria

Assessment

The progressive assessment will be carried out based in the following components:

- Exercises proposed for each course lesson
- A final project consisting of use-case building around the selected platform

The course grade will be composed of the following elements:

- Assessment of the proposed exercises: 15% (1.5 points)
- Assessment of the final project: 85% (8.5 points)

The assessment "Lesson 4: Project initial validation" will be held during the development of the final project. If this assessment is failed, it can be re-evaluated in the final test of the project.

Resit ("Examen Extraordinario") assessment

The resit assesment will consist of the following components:

- Course Project Exam: it is a written and practice exam
- Oral exam about the exercises proposed along the course

The grade will be composed of the following elements:

- Course Project: 80% (8 points)
- Oral exam: 20% (2 points)

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Zephyr RTOS Embedded C Programming: Using Embedded RTOS POSIX API (2024). Andrew Elias	Bibliography	
Zephyr Project Official Documentation	Web resource	Operating System Documentation. Available at: https://docs.zephyrproject.org/latest/index.html
Zephyr Project Website	Web resource	Zephyr Project Webpage. Hosted by Linux foundation. Available at: https://www.zephyrproject.org/

VS Code	Equipment	IDE Software. Available at: https://code.visualstudio.com/download
IoT Platform	Equipment	
UPM Moodle site	Web resource	
ESP32 Technical Reference Manual (TRM). Espressif Systems	Bibliography	Available at: https://www.espressif.com/sites/default/files/documentation/esp32_technical_reference_manual_en.pdf
Xtensa Instruction Set Architecture (ISA) Reference Manual	Bibliography	Available at: https://0x04.net/~mwk/doc/xtensa.pdf
Practical Electronics for Inventors (2016). Paul Scherz, Simon Monk	Bibliography	
The Designer's Guide to the Cortex-M Processor Family (2016 Second Edition) Trevor Martin	Bibliography	

9. Other information

9.1. Other information about the subject

The subject is related to Sustainable Development Goals number 9 and 12.

The deployment of IoT applications directly impacts the development of new industrial applications, improving the quality of life to minimise the environmental impact. In this course, we focus on developing IoT embedded systems that minimised the use of resources and energy consumption.

Due to the deprecation of Mbed OS in July 2026, the course will now introduce the use of Zephyr OS supported by the Linux Foundation as a broadly adopted alternative to the aforementioned operating system.